

Exp: 7

16 - Bit Multiplication

AIM:

To write an assembly language program to implement 16-bit multiplication using 8085 processor.

ALGORITHM:

- 1) Load the first data in HL pair
- 2) Move content of HL pair to stack pointer.
- 3) Load the second data in HL pair and move it to DE.
- 4) Make H register as 00H and L register as 00H.
- 5) ADD HL pair and stack pointer.
- 6) Check for carry if carry increment it by 1 else move to next step.
- 7) Then move E to A and perform OR operation with A and register D.
- 8) The value of operation is zero, then store the value to go to step.

PROGRAM:

MNEMONICS	EXPLANATION
LHLD 2050	loads content of 2050H and 2051H into the HL register pair.

SPHL Sets the stack pointer (SP) to the value in the HL register pair.

LHLD 2052 Loads the contents of 2052H and 2053H into the HL register pair.

XCHG Exchanges contents of HL and DE register pairs.

LXI H, 0000H Loads the value 0000H into the HL register pair.

LXI B, 0000H Loads the value 0000H into BC register pair.

AGAIN: DAD SP Marks the beginning of a loop, adds the content of SP register pair to HL register pair.

JNC START Jumps to start label if the carry flag is not set (i.e., no overflow occurred in the previous operation).

INX B Increments BC register pair by 1.

START: DCX D Marks the start of another loop, Decrement the DE register pair by 1.

MOV A, E Moves the contents of register E into register A.

ORA D Performs logical OR operation between A and D.

JNZ AGAIN Jumps back to AGAIN label if the zero flag is not set (i.e., if result of OR operation is non-zero).

SHLD 2054

Stores contents of the HL register into 2054H + 2055H.

MOV L, C

Moves the contents of C into L.

MOV H, B

Moves the contents of B into H.

SHLD 2055

Stores the contents of HL register pair into memory locations 2055H and 2056H.

HLT

Halts the program

INPUT:

Address	Data
2050	10
2052	5

0187

Output:

Address	Data
2054	50
2055	5

2

RESULT:

This the program was executed successfully using 6085 processor simulator.



Registers

A	00
BC	00 00
DE	00 00
HL	00 00
PSW	00 00
PC	42 22
SP	00 0A
Int-Reg	00

Flag

S	0
Z	1
AC	0
P	1
C	0

Load me at

```
1  LHLD 2050
2  SPHL
3  LHLD 2052
4  XCHG
5  LXI H,0000H
6  LXI B,0000H
7  AGAIN: DAD SP
8  JNC START
9  INX B
10 START: DCX D
11 MOV A,E
12 ORA D
13 JNZ AGAIN
14 SHLD 2054
15 MOV L,C
16 MOV H,B
17 SHLD 2055
18 HLT
```

Decimal - Hex Conversion

Decimal Hex

0	0
→ To Hex	← To Dec

I/O Ports

0	-	+	00
Update Port Value			

Memory

2052	-	+	05
Update Memory			

Data Stack KeyPad Memory I/O Ports

Start 2054

OK

Address (Hex)	Address	Data
0806	2054	50
0807	2055	0
0808	2056	0
0809	2057	0
080A	2058	0
080B	2059	0
080C	2060	0
080D	2061	0
080E	2062	0
080F	2063	0
0810	2064	0
0811	2065	0
0812	2066	0
0813	2067	0
0814	2068	0
0815	2069	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle

